

EN2014 Electronic Circuits and Analysis

Topic 01 - Biasing of Bipolar Junction Transistors

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Outline

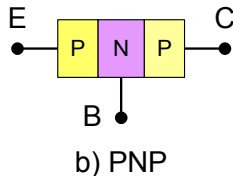
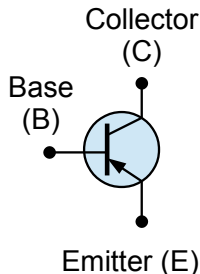
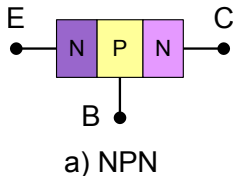
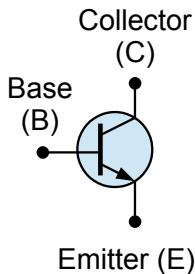
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- ➋ BJT Parameters
- ➌ Biasing Circuits
- ➍ Circuit Applications
 - Constant Current Sources
 - Current Mirrors
 - Current Multiplication
- ➎ Conclusion

Introduction

Bipolar Junction Transistors

- A three layer (two P-N junction) device
 - ▶ Heavily doped emitter
 - ▶ Lightly doped base
 - ▶ Moderately doped collector
- Can be NPN or PNP
 - ▶ Due to the higher mobility of electrons, NPN transistors are more widely used (specially for high frequency applications)
- The two P-N junctions have unique characteristics depending on the bias i.e., operating conditions of the device
 - ▶ Can be used as a switch or amplifier due to this

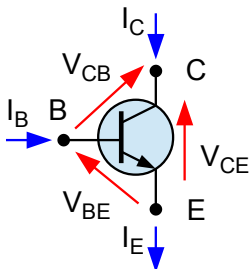
Transistor Symbols



Bipolar Junction Transistor Parameters

Bipolar Junction Transistor Biasing

- Biasing is the purposeful setting of operating conditions of a BJT in-order to obtain a specific behavior



V_{BE}	V_{CB}	Effect
RB	RB	Cut-off
FB	RB	Amplification
FB	FB	Saturation

Bipolar Junction Transistor Models

- Ebers-Moll model (1954)
- Gummel-Poon model (1970)
- Giacoletto (Hybrid- π) model (1969)
- Simplified models
 - ▶ T-model (M. V. Joyce and K. K. Clarke, Transistor Circuit Analysis. Reading, Mass.: Addison-Wesley, 1961, p. 39.)
 - ▶ h-parameter model (G. J. Pridham, Electronic Devices and Circuits. Pergammon Press, 1968, pp. 158-173)

Parametric Uncertainty

- The biasing of a BJT depends upon many parameters
 - ▶ Out of which I_B is the most convenient to control
- Some of these parameters may have significant uncertainty and therefore effect the device bias
 - ▶ β can vary due to fabrication errors and temperature
 - ▶ V_{BE} depends on temperature
- Biasing circuits should be less sensitive to such variations

Device Failure Modes

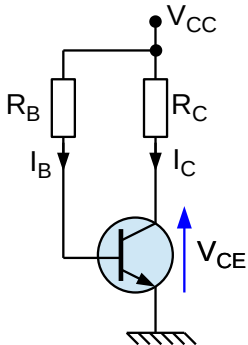
- The designed bias circuit has to be within the safe operating limits of the BJT
 - ▶ The “maximum ratings” of the device from the datasheet
- Even if the maximum ratings are not exceeded, a close value for a prolonged time can reduce the lifetime of the device
- Some failures can be reversed
- The best course of action is to avoid failures
 - ▶ For example thermal runaway
 - ▶ Requires failsafe circuit design

Biasing Circuits

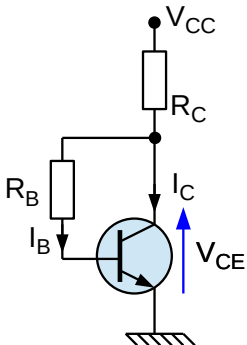
Overview

- A biasing circuit must provide the necessary operating conditions to BJT to obtain the intended purpose
- The robustness of a circuit has to be analyzed in terms of sensitivity to specific parameters
 - ▶ For a BJT β can be the most problematic
- The analysis can be analytic or empirical involving simulations
- Involves the dependance of I_C and V_{CE} on β

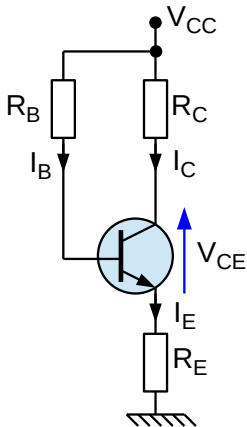
Fixed Bias



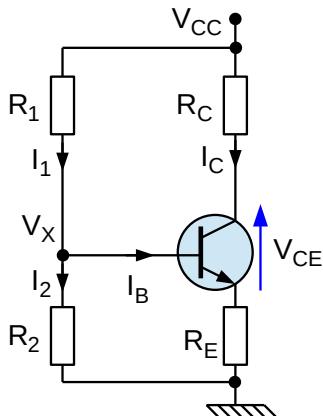
Collector Feedback Bias



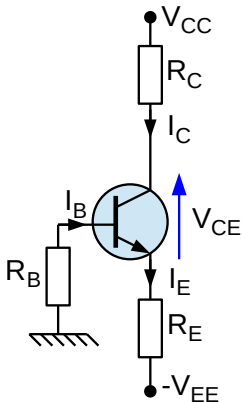
Fixed Bias with Emitter Resistor



Voltage Divider Bias



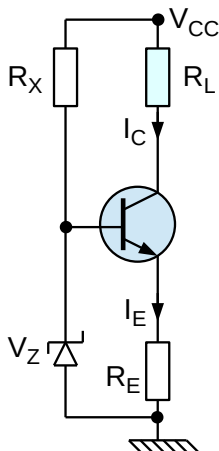
Emitter Bias



Circuit Applications

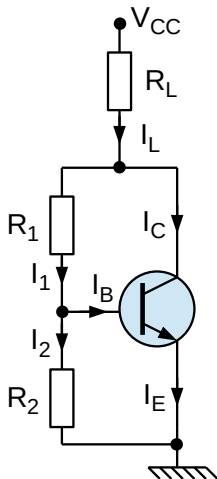
Constant Current Sources

Basic Constant Current Source



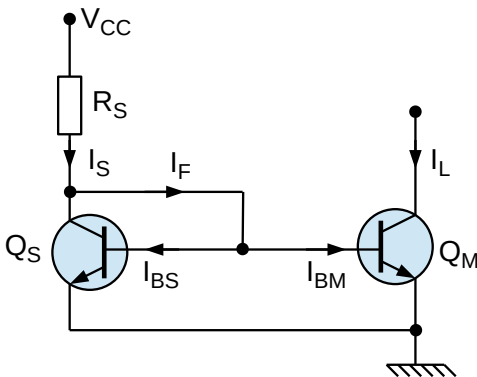
Constant Current Sources

Adjustable Voltage Clamp



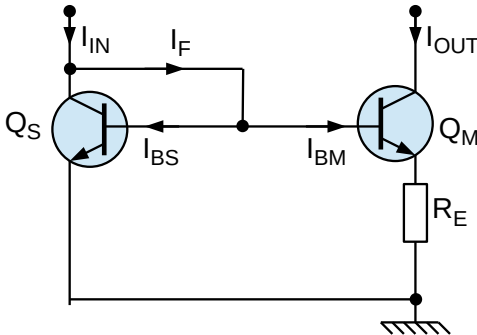
Current Mirrors

Basic Current Mirror



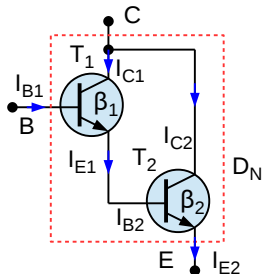
Current Mirrors

Widlar Current Mirror

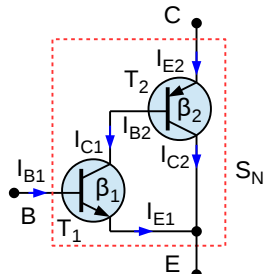


Current Multiplication

Current Multiplication



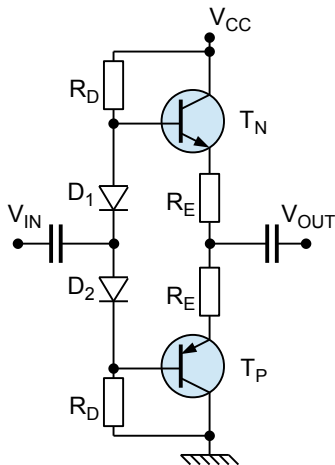
Darlington Configuration



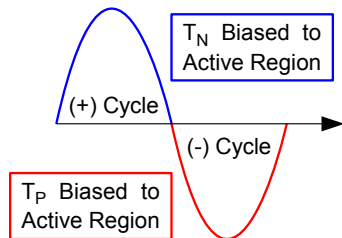
Sziklai Configuration

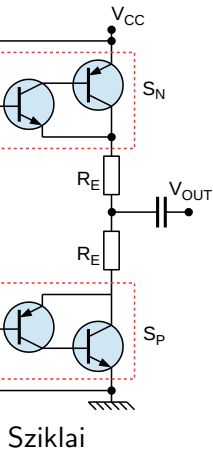
Current Multiplication

Class AB Power Amplifier



- R_E prevents thermal runaway
- Diode AC resistance is considered negligible
- $R_D \approx 1k\Omega$





Conclusion

Summary

- Biasing is the purposeful setting of operating conditions of a BJT in-order to obtain a specific behavior
- The biasing circuit has to be designed such that it is robust to device sensitive parameters
 - ▶ The most variable parameter for a BJT is β
- The voltage divider bias can be considered the most effective biasing circuit with a single supply